

risk, and insight into how the algorithm will behave in times of changing prices or varying market conditions. This transparency also allows investors to evaluate alternative strategies (algorithms) and determine the most appropriate “optimal” strategy (algorithm) given the underlying trading goal and investment objective.

HIGH FREQUENCY TRADING

High frequency trading (HFT) is the usage of sophisticated mathematical techniques and high speed computers to trade stocks, bonds, or options with the goal to earn a profit. This differs from the execution trading algorithms that are tasked with implementing an investment decision that has previously been determined. In other words, the HFT system makes both the investment and trading decisions simultaneously. High frequency trading in this sense is also called “black box” and “robo” trading.

HFT strategies can be classified into three different styles: Auto Market Making (AMM), Quant Trading/Statistical Arbitrage, and Rebate/Liquidity Trading. Donefer (2010) provides a similar classification in “Algos Gone Wild,” *Journal of Trading* (Spring 2010). There is often some overlap across these styles as we point out below, but for the most part, each of these styles has completely different goals and objectives. In short, high frequency trading has these features:

- *Automated trading.* Algorithms determine what to buy and what to sell, as well as the micro order placement strategies such as price, size, and timing of the trade. These decisions are determined from actual real-time market data including price signals, momentum, index or sector movement, volatility, liquidity, and order book information. These decisions are made independent of human interaction.
- *No net investment.* HFT does not require a large cash inflow since the inventory imbalances are netted out by the close each day. HFT strategies take both long and short positions in different names and close these position before the end of day so that they do not take on any overnight risk. In cases where the HFT holds overnight positions they will mostly likely use the proceeds from short sales to pay for the buys.
- *Short trading horizons.* Depending upon the strategies, HFT time horizons can vary from seconds to minutes, but also up to hours.

Auto Market Making

Auto market making (AMM) provides the financial community with the same services as the traditional market makers or specialists. The main